

Dongwook Kwon

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Experience

Suresoft Technologies

AI Engineer, Intelligent Agent Technology Team

Dec. 2025 – Present

Seongnam, Gyeonggi, Korea

- Developing an AI validation system utilizing AI agent-based architectures.
- Designing and implementing systems to verify and evaluate AI models through autonomous and collaborative AI agents.
- Building agentic AI systems to enable systematic, scalable, and reliable AI verification processes.

Bio-Computing and Machine Learning Laboratory, Kwangwoon University

Research Assistant

Aug. 2021 – Dec. 2025

Seoul, Korea

- Conducted research on deep learning algorithms, focusing on anomaly detection in time-series data, including Electrocardiogram (ECG) and Cyber-Physical Systems (CPS).
- Participated in the development of an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.

LG Electronics Canada

Research Project

Jan. 2025 – Jul. 2025

Toronto, ON, Canada

- Participating in the development of an assessment framework for agentic AI systems utilizing LLMs to evaluate performance.

Qualcomm Institute, University of California San Diego

AI Development Project, Summer Program

Jul. 2022 – Aug. 2022

San Diego, CA, USA

- Participated in the Personality-Based Drug Addiction Prediction System project.
- Attended seminars on deep learning and machine learning algorithms.

Education

Kwangwoon University

Master of Science in Computer Engineering

Mar. 2024 – Feb. 2026

Seoul, Korea

- GPA: 4.33/4.5 (Percentage: 98.1%)

University of Toronto

Graduate Research Program in Mechanical and Industrial Engineering

Jan. 2025 – Jul. 2025

Toronto, ON, Canada

- GPA: 3.68/4.0

Kwangwoon University*Bachelor of Science in Electronics and Communications Engineering*

Mar. 2019 – Feb. 2024

Seoul, Korea

- GPA: 3.41/4.5 (Percentage: 88.1%)

Honors & Awards

Major Awards

Aug. 2022 **Achievement Award**, Qualcomm Institute AI Development Project – UC San Diego *United States*

Dec. 2021 **Gold Award (Ministerial Award)**, 2021 Smart Maritime Logistics Project – Minister of Oceans and Fisheries *South Korea*

Academic Recognition

Nov. 2025 **Outstanding Student Paper Award**, Conference of the Institute of Electronics and Information Engineers *South Korea*

Nov. 2024 **Best Poster Award (Silver)**, 10th International Biomedical Engineering Conference (IBEC 2024) *South Korea*

Nov. 2023 **Outstanding Paper Award**, Conference of the Korean Society of Medical and Biological Engineering *South Korea*

Competitions

Dec. 2025 **Silver Award**, 23rd Embedded Software Competition (KIISE) *South Korea*

Dec. 2022 **Bronze Award**, Hanium ICT Mentoring Competition – FKII *South Korea*

Sep. 2022 **Excellence Award**, 18th Kwangwoon ICT Exhibition (KWIX) – Kwangwoon University *South Korea*

Oct. 2021 **Grand Prize**, MY (Multi-Y) Capstone Design Competition – Kwangwoon University *South Korea*

Funded Research Projects

Agentic AI Evaluation Framework*LG Electronics Canada*

Jan. 2025 – Present

Toronto, ON, Canada

- Developed an evaluation framework for agentic AI systems using LLMs to assess performance.

AI-based Intrusion Detection and Attack Packet Crafting Technologies for APR1400*Sejong University*

Jul. 2022 – Present

Seoul, Korea

- Developed an AI defense system to protect nuclear power plants from cyber attacks.

Quadruped Robot Using Vision SLAM with a Depth Camera*Ministry of Science and ICT*

Apr. 2022 – Nov. 2022

Seoul, Korea

- Developed an autonomous quadruped robot using a navigation algorithm with SLAM and a depth camera.

Personality-Based Drug Addiction Prediction System

Qualcomm Institute, UC San Diego

Jul. 2022 – Aug. 2022

San Diego, CA, USA

- Developed a personality-based drug addiction prediction system using a machine learning model.

Active Kill-Switch and Biomarker-Based Defense System for Life-Threatening IoT Medical Devices

Kookmin University

Sep. 2021 – Dec. 2022

Seoul, Korea

- Developed a security system for heart implant devices using an anomaly detection algorithm.

Smart Logistics System Using Computer Vision

Ministry of Oceans and Fisheries

Apr. 2021 – Nov. 2021

Seoul, Korea

- Developed a logistics detection system incorporating location tracking and stack-layer identification for containers.

Smart Kitchen IoT System Utilizing Computer Vision and Deep Learning

Kwangwoon University

May 2021 – Oct. 2021

Seoul, Korea

- Developed a cloud IoT platform featuring a system for classifying groceries in the fridge and an OCR receipt program.

Publications

Journal Articles

- Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, “Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise,” *Electronics*, vol. 14, no. 5, p. 890, 2025.
- **D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, “Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments,” under review at *Data Mining and Knowledge Discovery* (Springer Nature), 2025.

Conference Papers

- R. Chang[†], **D. Kwon**[†], J. Lee[†], and N. Verma, “CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades,” in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) – Industry Track (Poster)*, 2026. [†]Equal contribution.
- J. Lee[†], R. Chang[†], **D. Kwon**[†], H. Singh, and N. Verma, “GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems,” in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. [†]Equal contribution.
- **D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, “Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers,” in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award – Silver.**
- **D. Kwon**, Y. Kang, and C. Park, “Enhancing medical device security with GNN-GRU anomaly detection model,” in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper**

Award.

Patents

- **Issued Patent**, “Method for Implantable Medical Device to Detect Anomaly in Real Time,” Patent No. KR 10-2961984, Issued on April 30, 2026.
- **Issued Patent**, “GPT API-Based Network Anomaly Detection,” Patent No. KR 10-2848814, Issued on August 18, 2025.

Leadership Experience

- **President**, Kwangwoon International Student Association – Lead intercultural programs and student engagement initiatives.
- **Director of Filming and Editing**, Kwangwoon Broadcasting Center (KWBC) – Oversaw media production and content delivery.
- **Class Representative**, Dept. of Electronics and Communications Engineering – Represented student interests in academic affairs.
- **Executive Member**, Amateur Radio Club – Organized radio workshops and technical exhibitions.